AI-Powered Interior Redesign System

Image-to-Image Generation with Custom Fine-Tuned FLUX.2 Klein (4B) via 4-Bit QLoRA

Student Author:	Lokesh Chakraborty	**Local Hardware:**	ASUS Vivobook 16 (i5-12500H, Iris Xe)
Model Architecture:	FLUX.2 Klein Base 4B (DiT)	**Fine-Tuning Method:**	4-Bit QLoRA (NF4) via PEFT
Deployment Platform:	Hugging Face ZeroGPU (A100)	**Local Client Footprint:**	0 MB VRAM, ~3.5s per image

1. Executive Summary & Problem Formulation

Modern architectural and interior visualization typically requires commercial CAD software, manual 3D rendering, or enterprise-grade cloud compute. The objective of this project was to construct an autonomous, end-to-end **AI-driven Interior Redesign System** where an end user uploads an image of an unadorned, empty, or outdated room and receives a high-resolution (1024×1024), photorealistic redesigned room matching a custom luxury aesthetic.

The Core Engineering Challenge: The system had to be developed and operated using a standard consumer ultrabook (ASUS Vivobook 16 equipped with an Intel Core i5-12500H processor and integrated Intel Iris Xe Graphics). Because integrated GPUs have **0 MB of dedicated Video RAM (VRAM)** and lack NVIDIA CUDA acceleration, running or training modern generative models locally was computationally impossible without system freezing or Out-Of-Memory (OOM) failures. This project successfully bypassed hardware boundaries by combining **Quantized Low-Rank Adaptation (4-bit QLoRA)** in the cloud, automated dataset curation, and a serverless **Distributed Cloud Inference Client**.

2. Architectural Evaluation & Model Selection

During research and prototyping, multiple generative diffusion backbones were experimentally tested to determine which model could best preserve room geometry while synthesizing realistic furniture, lighting, and textures.

Model Architecture	Model Size	VRAM Req.	Interior Fidelity & Geometry Assessment
Stable Diffusion 1.5	~2.1 GB	4 GB - 6 GB	High speed; but struggles with architectural straight lines and prompt fidelity without ControlNet.
Realistic Vision v5.1	~4.2 GB	6 GB - 8 GB	Photorealistic textures, but prone to warping wall boundaries in direct img2img mode.
FLUX.2 Klein Base (4B) ***(Selected System)***	~8.2 GB (16-bit) ~2.2 GB (4-bit)	~4.5 GB (QLoRA)	**State-of-the-Art**: Rectified Flow Transformer (MMDiT) with native multi-reference editing and Qwen text encoder.

3. Dataset Engineering & Automated Curation Pipeline

Supervised fine-tuning of generative models requires strict image-caption consistency. A customized data-preparation script (prepare_dataset.py) was developed to autonomously stream, filter, format, and package 50 ultra-high-resolution indoor living space pairs from Hugging Face datasets.

Pipeline Execution Stages:

- **Streaming Ingestion:** Image streaming from MohamedAli77/interior-rooms and ellljoy/interior-design.
- **Smart Center-Cropping:** Automatically cropped variable aspect-ratio photographs into strict 1:1 squares using Lanczos resampling to prevent distortion.
- **Resolution Normalization:** Upscaled/standardized to native 1024×1024 pixel tensors matching FLUX patch requirements.
- **Trigger Word Synthesis:** Every caption was prepended with a custom trigger keyword: modern_interior (e.g. *'modern_interior, living room with white marble floors, white oak coffee table, and warm sunlight'*).
- **Packaging:** Automatically compressed into a 12.1 MB portable archive (interior_lora_dataset.zip) for cloud ingestion.

4. Mathematical Foundation & 4-Bit QLoRA Fine-Tuning

Full model fine-tuning of a 4-Billion parameter transformer requires computing gradients and optimizer states for all weights, demanding over **45 GB to 60 GB of VRAM** (far exceeding any free GPU tier). To overcome this, the project implemented **Quantized Low-Rank Adaptation (QLoRA)**.

Mathematical Principle of LoRA:

During LoRA adaptation, the pre-trained weight matrix $W_0 \in \blacksquare^{d \times k}$ is frozen and does not receive gradient updates. Instead, parameter updates are factorized into low-rank decomposition matrices $A \in \blacksquare^{r \times k}$ and $B \in \blacksquare^{d \times r}$, where rank $r \blacksquare min(d, k)$:

$$W = W_0 + \Delta W = W_0 + (\alpha / r) \cdot (B \cdot A)$$

Where rank r = 16, scaling factor α = 16, and target projection layers = {to_q, to_k, to_v, to_out}.

Parameter Efficiency Analysis:

During training initialization, the PEFT wrapper confirmed the following parameter reduction:

Metric	Full FLUX.2 Base 4B	Our Custom LoRA Adapter	Efficiency Ratio
Trainable Parameters	3,877,510,656 params	**1,966,080 params**	**99.949% Frozen** (Only 0.05% trained!)
Disk / Memory Size	~8.2 GB - 16.0 GB	**8.37 MB (.safetensors)**	**1,000× smaller** file footprint

Engineering Breakthroughs in the Training Pipeline:

- **4-Bit NormalFloat4 (NF4) Quantization:** Base DiT weights were quantized to 4-bit using bitsandbytes, shrinking baseline memory from 8.5 GB down to 2.2 GB VRAM.

- **Latent Pre-Caching (--cache_latents):** All 50 training images were encoded through the VAE once upfront. The VAE model was then unloaded from GPU memory, completely eliminating PCIe memory-swapping bottlenecks and dropping epoch time from hours to minutes.
- **8-Bit Adam Optimizer (AdamW8bit):** Reduced optimizer memory state from 4 GB down to ~50 MB.
- **Gradient Checkpointing:** Recomputed activations during the backward pass to maintain low peak VRAM headroom under 5.0 GB on Google Colab's single NVIDIA T4 GPU.

5. Distributed Client-Server Deployment Architecture

To allow the local ASUS Vivobook laptop to redesign rooms with zero local GPU compute, a decoupled client-server architecture was established:

Local Client (Vivobook)	Secure API Gateway	Serverless Cloud Server (Hugging Face)
• User selects local room photo (my_room.jpg) • Formulates style prompt (modern_interior, ...) • Runs lightweight Python script (text_redisgn.py) • **Hardware:** 0 MB VRAM, ~2% CPU usage	■■(HTTPS POST)■■> Gradio Client RPC Auto-authentication <■■(JPEG Response)■	• **Hardware:** NVIDIA A100 GPU (ZeroGPU) • **Core Engine:** FLUX.2-klein-4B (BF16) • **Custom Adapter:** pytorch_lora_weights.safetensors • **Latency:** ~3.5 seconds generation time • **Output:** 1024×1024 Photorealistic Render

6. Experimental Results & Verification

The system was validated through live image-to-image inference runs using test photographs of unrenovated rooms. The fine-tuned FLUX.2 Klein model demonstrated remarkable performance across three key criteria:

- **Structural Integrity:** Room perspective, ceiling heights, window placement, and wall orientations from the user's uploaded image were accurately retained.
- **Material Realism:** The fine-tuned weights successfully introduced high-end Scandinavian finishes including natural light oak hardwood, neutral linen upholstery, marble coffee tables, and organic indoor flora.
- **Inference Efficiency:** While generating locally would have taken over 20 minutes per image on an Intel CPU, the distributed ZeroGPU pipeline returned verified 1024×1024 JPEG results (flux_redesigned_room.jpg) in **under 4 seconds**.

7. Summary of Delivered Project Artifacts

Artifact Name	Type / Location	Technical Purpose & Role
`prepare_dataset.py`	Python Script	Streams HF datasets, crops to 1024×1024, injects trigger word, and archives pairs.
`interior_lora_dataset.zip`	Dataset Archive (12.1 MB)	50 curated, captioned, and formatted image-caption training pairs.
`pytorch_lora_weights.safetensors`	LoRA Model (8.37 MB)	Trained low-rank adapter containing 1.96M interior design weights.
`app.py (Hugging Face Space)`	Cloud Serverless Backend	ZeroGPU Gradio endpoint serving FLUX.2 Klein + fine-tuned LoRA on A100.
`text_redisgn.py`	Local Client Script	Sends local room photos to cloud and downloads 1024×1024 results in seconds.
`flux_redesigned_room.jpg`	Output Artifact (250 KB)	Live verified 1024×1024 redesigned interior room photograph.

Conclusion & Academic Assessment:
This project proves that cutting-edge Generative AI models can be adapted and deployed without high-budget local hardware. Through mathematical parameter-efficiency (QLoRA) and cloud-native architecture, enterprise-grade interior architectural design was achieved on a lightweight student laptop.